MC-ARM

Team Emertxe

DAY 2

Today

- What are we going to do today ?
  Write our first code to Blink Led
- Where ?
- How ?

Important Terms

- Host: A system which is used to develop the target.

  Laptop

  IDE :- STM32 cube IDE

- Target:A system which is being developed for specific application.

  Simulator:-

  PICSIMLAB => board -> Blue pill => Micro controller -> stm32f103c8t6

Section 1 – A, R and M profiles - Cortex-M Series

ARM Processor Families

What is ARM?

- ARM = Advanced RISC Machine
- RISC-based processor architecture
- Low power, high performance
- Widely used in embedded systems, mobile devices, IoT
- ARM designs CPU cores; companies manufacture chips (ST, NXP, TI, Microchip)

95%+
of all smartphones run
on ARM cores

200+
silicon vendors license
ARM architecture

30B+
ARM-based chips shipped
in 2023 alone

ARM Processor Profiles

Application
Cortex-A

High-performance systems
Runs Linux, Android, iOS

Smartphones
Tablets
Raspberry Pi

Real-Time
Cortex-R

Real-time & safety-critical
Deterministic response

Automotive ECUs
Industrial automation
Medical devices

Microcontroller
Cortex-M

Microcontrollers
Low power, low cost

IoT devices
Sensors
Embedded control

Cortex-M Series Overview

Cortex-M0 / M0+
Smallest, most energy efficient
For ultra-low-power IoT

Cortex-M3
General-purpose
No DSP / FPU — STM32F1

Cortex-M4
DSP + optional FPU
Signal processing — STM32F4

Cortex-M7
High performance
Dual issue, large cache

Increasing Performance, Features, and Power →

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STM32 Family Overview

- STM32 is ARM Cortex-based MCU family by STMicroelectronics
- Series :- STM32F , STM32 L , STM32H , SMT32G , STM32W

STM32F1 Series

Cortex-M3 core — up to 72 MHz
Low cost, basic peripherals
Best for: LED, sensors, communication

WAP to blink LED

- Where LED is connected to ?

WAP to blink LED

- Total 48 pins

  PORTA - 16 pins [PA0 - PA15]

  PORTB - 16 pins [PB0 - PB15]

  PORTC - 3 pins [PC13 - PC15]

  PORTD - 2 pins [PD 0 , PD1]

  total   = 37 I/O

HAL functions

Commonly used HAL functions in STM32 programing

To assign the value to the pin

- HAL_GPIO_WritePin(PORT , PIN_NO , VALUE);

  ex :

  ```
  HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
  HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
  ```

- HAL_GPIO_TogglePin(GPIOx , GPIO_Pin);

  ex:HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);

WAP to blink LED

```
while (1)
{
HAL_GPIO_TogglePin(GPIOC, GPIO_PIN_13);
HAL_Delay(500);
}
```

THANK YOU

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